In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
data=pd.read_csv("D:/Pg_data.csv")
```

In [4]:
```
data.head()
```

Out[4]:

	order_ID	Order_Date	Customer ID	Customer_Name	Gender	Age	Region	City
0	ORD-1001	01-01-25	CUST-1001	Naveed	Male	30	Sindh	Karachi
1	ORD-1002	02-01-25	CUST-1002	Umar	FeMale	32	Punjab	Lahore
2	ORD-1003	03-01-25	CUST-1003	Ishaq	Male	39	Sindh	Karachi
3	ORD-1004	04-01-25	CUST-1004	Abas	Male	29	balochistan	Quetta
4	ORD-1005	05-01-25	CUST-1005	Adnan	FeMale	27	Khyber Pakhtunkhwa	peshawar

In [11]:
```
data['gross_sales']=data['Quantity'] * data['Unit_Price']
data['discount_amount']=data['gross_sales'] * (data['Discount']/ 100.0)
data['net_sales']=data['gross_sales'] - data['discount_amount']
data['total_cost'] = data['Quantity'] * data['Unit_Cost']
data['profit'] = data['net_sales'] - data['total_cost']
data['profit_margin'] = np.where(data['net_sales'] > 0 ,data['profit'] / data['net_sa
```

In [13]:
```
data.columns
```

Out[13]:
```
Index(['order_ID', 'Order_Date', 'Customer ID', 'Customer_Name', 'Gender',
       'Age', 'Region', 'City', 'Sales_Channel', 'Product_Category', 'Product',
       'Quantity', 'Unit_Price', 'Unit_Cost', 'Discount', 'Payment_Method',
       'Salesperson', 'Customer_Rating', 'gross_sales', 'discount_amount',
       'net_sales', 'total_cost', 'profit', 'profit_margin'],
      dtype='str')
```

In [14]:
```
data.head()
```

Out[14]:

	order_ID	Order_Date	Customer ID	Customer_Name	Gender	Age	Region	City
0	ORD-1001	01-01-25	CUST-1001	Naveed	Male	30	Sindh	Karachi
1	ORD-1002	02-01-25	CUST-1002	Umar	FeMale	32	Punjab	Lahore
2	ORD-1003	03-01-25	CUST-1003	Ishaq	Male	39	Sindh	Karachi
3	ORD-1004	04-01-25	CUST-1004	Abas	Male	29	balochistan	Quetta
4	ORD-1005	05-01-25	CUST-1005	Adnan	FeMale	27	Khyber Pakhtunkhwa	peshawar

5 rows × 24 columns

In [17]:
```
col_round=['gross_sales',       'discount_amount',      'net_sales',    'total_cost',
data[col_round]=data[col_round].round(1)
data['profit_margin'] = data['profit_margin'].round(4)
```

In [19]:
```
data.tail()
```

Out[19]:

	order_ID	Order_Date	Customer ID	Customer_Name	Gender	Age	Region	City
95	ORD-1096	06-04-25	CUST-1096	Habib	FeMale	31	Sindh	Karach
96	ORD-1097	07-04-25	CUST-1097	Naveed	Male	35	punjab	Lahore
97	ORD-1098	08-04-25	CUST-1098	Kaleem	Male	25	Sindh	Karach
98	ORD-1099	09-04-25	CUST-1099	Ishaq	Male	33	balochistan	Quetta
99	ORD-1100	10-04-25	CUST-1100	Adnan	FeMale	26	Khyber Pakhtunkhwa	peshawa

5 rows × 24 columns

In [20]:
```
data.isnull().sum()
```

Out[20]:
```
order_ID            0
Order_Date          0
Customer ID         0
Customer_Name       0
Gender              0
Age                 0
Region              0
City                0
Sales_Channel       0
Product_Category    0
Product             0
Quantity            0
Unit_Price          0
Unit_Cost           0
Discount            0
Payment_Method      0
Salesperson         0
Customer_Rating     0
gross_sales         0
discount_amount     0
net_sales           0
total_cost          0
profit              0
profit_margin       0
dtype: int64
```

In [21]:
```
data.describe()
```

Out[21]:

	Age	Quantity	Unit_Price	Unit_Cost	Discount	Customer_Rating	
count	100.000000	100.000000	100.00000	100.00000	100.000000	100.000000	
mean	32.250000	3.170000	13625.00000	12095.00000	8.560000	2.680000	47
std	4.710562	1.442886	21110.93534	19097.15921	4.484136	1.406191	81
min	25.000000	1.000000	800.00000	500.00000	0.000000	1.000000	
25%	28.000000	2.000000	1500.00000	1100.00000	5.000000	1.000000	3
50%	32.000000	3.000000	2000.00000	1600.00000	9.000000	3.000000	6
75%	36.000000	5.000000	14000.00000	12450.00000	13.000000	4.000000	20
max	40.000000	5.000000	50000.00000	45000.00000	15.000000	5.000000	250

In [22]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 24 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_ID          100 non-null    str
 1   Order_Date        100 non-null    str
 2   Customer ID       100 non-null    str
 3   Customer_Name     100 non-null    str
 4   Gender            100 non-null    str
 5   Age               100 non-null    int64
 6   Region            100 non-null    str
 7   City              100 non-null    str
 8   Sales_Channel     100 non-null    str
 9   Product_Category  100 non-null    str
 10  Product           100 non-null    str
 11  Quantity          100 non-null    int64
 12  Unit_Price        100 non-null    int64
 13  Unit_Cost         100 non-null    int64
 14  Discount          100 non-null    int64
 15  Payment_Method    100 non-null    str
 16  Salesperson       100 non-null    str
 17  Customer_Rating   100 non-null    int64
 18  gross_sales       100 non-null    int64
 19  discount_amount   100 non-null    float64
 20  net_sales         100 non-null    float64
 21  total_cost        100 non-null    int64
 22  profit            100 non-null    float64
 23  profit_margin     100 non-null    float64
dtypes: float64(4), int64(8), str(12)
memory usage: 18.9 KB
```

In [24]:
```
data.shape
```

Out[24]:
```
(100, 24)
```

In [34]:
```
data.columns
```

Out[34]:
```
Index(['order_ID', 'Order_Date', 'Customer ID', 'Customer_Name', 'Gender',
       'Age', 'Region', 'City', 'Sales_Channel', 'Product_Category', 'Product',
       'Quantity', 'Unit_Price', 'Unit_Cost', 'Discount', 'Payment_Method',
       'Salesperson', 'Customer_Rating', 'gross_sales', 'discount_amount',
       'net_sales', 'total_cost', 'profit', 'profit_margin'],
      dtype='str')
```

In [36]:
```
data.groupby('Order_Date')['net_sales'].sum()
```

Out[36]:
```
Order_Date
01-01-25      1504.0
01-02-25    184000.0
01-03-25      2040.0
01-04-25      2208.0
02-01-25      4185.0
              ...
29-03-25      3440.0
30-01-25      7120.0
30-03-25      6675.0
31-01-25      1425.0
31-03-25    200000.0
Name: net_sales, Length: 100, dtype: float64
```

In [43]:
```
# Step 1: Text column ko Datetime mein convert karein
data['Order_Date'] = pd.to_datetime(data['Order_Date'])

# Step 2: Ab year extract karein
data['year'] = data['Order_Date'].dt.year
```

In [49]:
```
data['month'] = data['Order_Date'].dt.month
data['month'] = data['Order_Date'].dt.strftime("%b")
```

In [50]:
```
data['day'] = data['Order_Date'].dt.day
```

In [51]:
```
data.columns
```

Out[51]:
```
Index(['order_ID', 'Order_Date', 'Customer ID', 'Customer_Name', 'Gender',
       'Age', 'Region', 'City', 'Sales_Channel', 'Product_Category', 'Product',
       'Quantity', 'Unit_Price', 'Unit_Cost', 'Discount', 'Payment_Method',
       'Salesperson', 'Customer_Rating', 'day', 'month', 'year', 'gross_sales',
       'discount_amount', 'net_sales', 'total_cost', 'profit',
       'profit_margin'],
      dtype='str')
```

In [52]:
```
new=['order_ID', 'Order_Date', 'Customer ID', 'Customer_Name', 'Gender',
       'Age', 'Region', 'City', 'Sales_Channel', 'Product_Category', 'Product',
       'Quantity', 'Unit_Price', 'Unit_Cost', 'Discount', 'Payment_Method',
       'Salesperson', 'Customer_Rating','day','month','year', 'gross_sales', 'discour
       'net_sales', 'total_cost', 'profit', 'profit_margin']
data=data[new]
```

In [48]:
```
data.head()
```

Out[48]:

	order_ID	Order_Date	Customer ID	Customer_Name	Gender	Age	Region	City
0	ORD-1001	2025-01-01	CUST-1001	Naveed	Male	30	Sindh	Karachi
1	ORD-1002	2025-02-01	CUST-1002	Umar	FeMale	32	Punjab	Lahore
2	ORD-1003	2025-03-01	CUST-1003	Ishaq	Male	39	Sindh	Karachi
3	ORD-1004	2025-04-01	CUST-1004	Abas	Male	29	balochistan	Quetta
4	ORD-1005	2025-05-01	CUST-1005	Adnan	FeMale	27	Khyber Pakhtunkhwa	peshawar

5 rows × 27 columns

In [55]:
```
data.groupby('month')['profit'].sum()
```

Out[55]:
```
month
Apr    -6300.0
Aug    -5076.0
Dec    -1385.0
Feb    30175.0
Jan    30311.0
Jul     3492.0
Jun     1956.0
Mar    31051.0
May    11950.0
Nov     9481.0
Oct     4195.0
Sep     2780.0
Name: profit, dtype: float64
```

In [61]:
```
new2= ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
               'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
data['month']=pd.Categorical(data['month'],categories=new2,ordered=True)
```

In [64]:
```
data['month'].value_counts().sort_index()
```

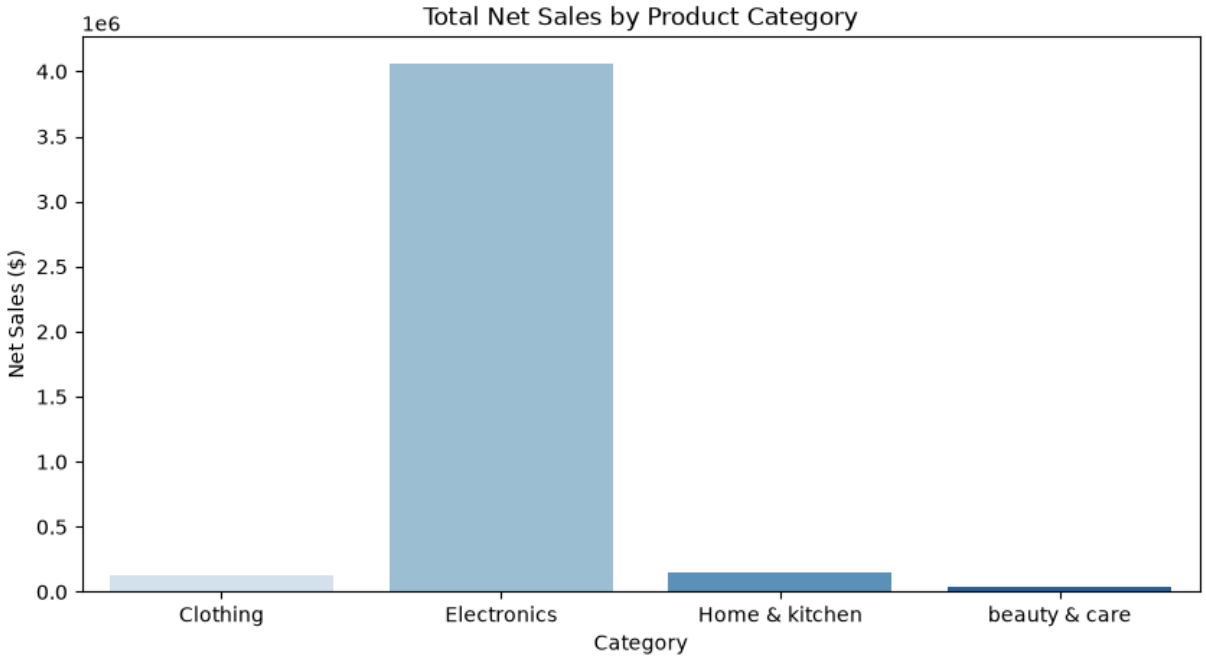
Out[64]:
```
month
Jan    23
Feb    20
Mar    23
Apr     4
May     4
Jun     4
Jul     4
Aug     4
Sep     4
Oct     4
Nov     3
Dec     3
Name: count, dtype: int64
```

In [76]:
```
# 1. Top Performing Product Categories (Bar Chart)
plt.figure(figsize=(10, 5))
category_sales = data.groupby('Product_Category')['net_sales'].sum().reset_index()
sns.barplot(data=category_sales, x='Product_Category', y='net_sales', palette='Blues'
plt.title('Total Net Sales by Product Category')
plt.xlabel('Category')
plt.ylabel('Net Sales ($)')
plt.show()
```

```
C:\Users\pc\AppData\Local\Temp\ipykernel_4876\2019907718.py:4: FutureWarning:

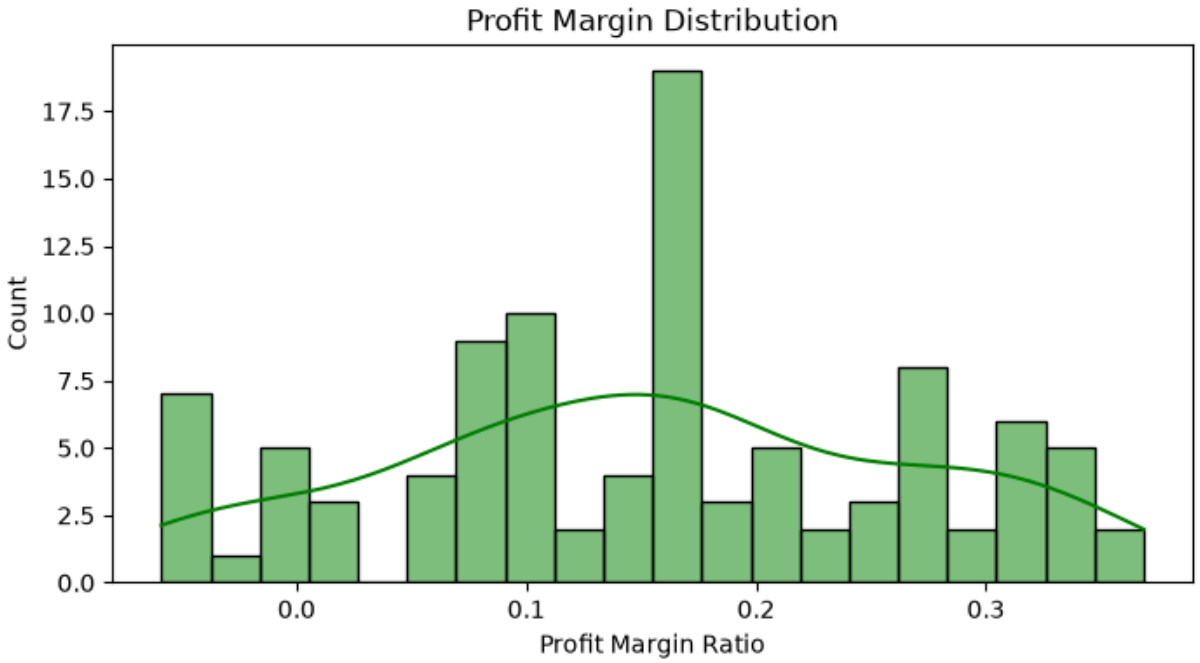
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.
0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=category_sales, x='Product_Category', y='net_sales', palette='Blue
s')
```



In [72]:
```
plt.figure(figsize=(8, 4))
sns.histplot(data['profit_margin'], bins=20, kde=True, color='green')
plt.title('Profit Margin Distribution')
```

```
plt.xlabel('Profit Margin Ratio')
plt.show()
```



In [75]:
```
data.to_excel('retail_sales_data.xlsx' , index=False)
```

In [ ]:
```

```